REGULATORY DOSSIER SUBMISSION
Dossier ID: DOS-FED2490BB8
Country: Burkina Faso
Submission Date: 2025-11-23
Product: Nitrvia (nitrofurantoin) | ATC: J01XE01 | Strength: 250 mg
Applicant: United Therapeutics | Manufacturer: Bobo-Dioulasso Pharma SA
Policy Label: reject_and_return | Risk Score: 0.92
AMR Stewardship: AWaRe=access | Unmet Need=routine | GLASS Trend=stable | Watch Similarity=low

SECTION CONTENT

[1] m1_application_admin - Application Form and Administrative Information
Labels: presence=present; length=length_ok; correctness=correct
This dossier constitutes a formal application for Marketing Authorization of Nitrvia (nitrofurantoin) submitted on 2025-11-23 to the national regulatory authority of Burkina Faso. The applicant, United Therapeutics, formally requests approval for the commercial distribution of this medicinal product. The proposed pharmaceutical form is capsule with a strength of 250 mg, classified under ATC code J01XE01. All required administrative declarations, legal attestations, letters of authorization, and regional application forms have been completed, signed by the responsible Qualified Person, and appended to Module 1. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The proposed prescribing information and patient information leaflet have been translated into local languages. The applicant confirms alignment with applicable technical guidance and regional submission standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. All referenced documents are available in the dossier annex and traceable by section identifier. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Pharma

[1] m1_manufacturer_gmp - Manufacturer and GMP Evidence
Labels: presence=present; length=length_ok; correctness=partial
Error tags: gmp_outdated
Latest GMP inspection date is 2020-12-01. No recent inspection evidence was provided for the required review window. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. All referenced documents are available in the dossier annex and traceable by section identifier. The applicant confirms alignment with applicable technical guidance and regional submission standards. Method validation records and quality controls were reviewed for internal consistency. The GMP evidence package includes certificate status, inspection scope, and manufacturing-site accountability records. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Batch manufacturing records and standard operating procedures (SOPs) are maintained and archived on-

site. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Recent regulatory inspections yielded no critical observations, and all major findings have been effectively closed. Site quality-system commitments were compared against the proposed commercial manufacturing activities. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Where applicable, protocol d

[1] m1_product_information - Product Information and Naming
Labels: presence=present; length=length_ok; correctness=correct
The proposed proprietary name for this medicinal product is Nitrvia, containing the active pharmaceutical ingredient nitrofurantoin. The draft Summary of Product Characteristics (SmPC), Patient Information Leaflet (PIL), and primary/secondary packaging labels are provided for review. These documents detail the approved therapeutic indications, posology, contraindications, special warnings, and precautions for use. A comprehensive risk management plan to minimize medication errors is also submitted. WHO AWaRe category: Access. Product is positioned for routine antibacterial use with standard stewardship monitoring and local resistance surveillance. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The applicant confirms alig

[2] m2_quality_overall_summary - Quality Overall Summary
Labels: presence=present; length=length_ok; correctness=correct
This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for nitrofurantoin. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. All referenced documents are available in the dossier annex and traceable by section identifier. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The applicant confirms alignment with applicable technical guidance and regional submission standards. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Method validation records and quality controls were reviewed for internal consistency. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. All referenced documents are available in the dossier annex and traceable by section identifier. Risk mitigation activities are documented with ownership, timelines, and

follow-up checkpoints. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Method validation records and quality controls we

[2] m2_clinical_overview - Clinical Overview and Benefit-Risk Summary
Labels: presence=present; length=length_ok; correctness=incorrect
Error tags: clinical_failed
Clinical overview for type 2 diabetes indicates pivotal efficacy outcomes did not meet the primary endpoint. Current evidence does not support a favorable benefit-risk conclusion for authorization. The target population demographics are broadly representative of the intended commercial patient cohort. No new safety signals were identified during the extended open-label follow-up period. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Method validation records and quality controls were reviewed for internal consistency. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Clinical interpretation focuses on whether efficacy, safety, and benefit-risk conclusions are supported by the submitted evidence. The clinical overview links the therapeutic rationale to the pivotal evidence package and the proposed indication. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The applicant confirms alignment with applicable technical

[3] m3_api_quality - API Quality and Control Strategy
Labels: presence=present; length=length_ok; correctness=correct
The active pharmaceutical ingredient (API) section details the complete synthesis route, starting from well-defined regulatory starting materials. Robust in-process controls and intermediate specifications ensure the consistent quality of the final drug substance. The analytical procedures used for release and stability testing have been fully validated for accuracy, precision, linearity, and robustness. The impurity control strategy adequately addresses organic impurities, residual solvents, and elemental impurities in accordance with current ICH guidelines. Forced degradation studies confirm the stability-indicating nature of the assay methods. The API package links specification limits to validated analytical methods and impurity-control decisions. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Residual solvents and elemental impurities are controlled well below ICH Q3C and Q3D permissible daily exposures. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Batch analysis, validation, and impurity management records support the proposed API quality framework. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The synthetic route involves three well-characterized steps with defined starting materials and isolated intermediates. The manufacturing site operates under a fully implemented Pharmaceutical Quality

System (PQS) aligned with ICH Q10. The applicant confirms alignment with applicable technical guidance and regional submission standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Method validation records and quality controls were reviewed for internal consistency. Pharmacokinetic parameters were

[3] m3_fpp_manufacturing - FPP Manufacturing Process and Controls
Labels: presence=present; length=length_ok; correctness=correct
The finished pharmaceutical product (FPP) manufacturing process utilizes standard, scalable unit operations. A formal quality risk assessment (e.g., FMEA) was conducted to identify critical process parameters (CPPs) that impact critical quality attributes (CQAs). Process performance qualification (PPQ) reports for three consecutive commercial-scale batches verify that the process is maintained in a state of control. In-process controls, intermediate hold-time studies, and packaging validation data are presented to substantiate the robustness and reproducibility of the commercial manufacturing operations. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Method validation records and quality controls were reviewed for internal consistency. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The applicant confirms alignment with applicable technical guidance and regional submission standards. All referenced documents are available in the dossier annex and traceable by section identifier. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. All referenced documents are available in the dossier annex and traceable by section identifier. Where applicable, protocol deviations were assessed for impact on interpretatio

[3] m3_stability - Stability and Shelf-Life Justification
Labels: presence=present; length=length_ok; correctness=correct
A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40℃/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation

trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Method validation records and quality controls were reviewed for internal consistency. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The applicant confirms alignment with applicable technical guidance and regional submission standards. All referenced documents are available in the dossier annex and traceable by section identifier. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Data were analyzed using a mixed-effects model for

[4] m4_nonclinical_summary - Nonclinical Study Summary
Labels: presence=present; length=length_ok; correctness=correct
The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound. Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Where applicable, protocol deviations were assessed for impact on inter

[5] m5_trial_listing - Tabular Listing of Clinical Studies
Labels: presence=present; length=length_ok; correctness=correct
The tabular listing of clinical studies provides a comprehensive inventory of all trials conducted in support of the proposed indication (type 2 diabetes). This includes Phase I pharmacokinetic/pharmacodynamic studies, Phase II dose-finding studies, and the pivotal Phase III efficacy and safety trials. For each study, the table details the protocol identifier, study design, randomization ratio, number of enrolled subjects, study duration, and current completion status. GLASS-aligned surveillance trend is stable, supporting continued first-line or second-line use with routine monitoring. All referenced documents are available in the dossier annex and traceable by section identifier. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Statistical significance was set at an alpha level of 0.05, and all tests were

two-sided. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The applicant confirms alignment with applicable technical

[5] m5_pivotal_trial_reports - Pivotal Clinical Trial Reports
Labels: presence=present; length=length_ok; correctness=incorrect
Error tags: clinical_failed
Pivotal trial analysis indicates primary endpoint was not met in the intent-to-treat population. Estimated treatment effect did not reach pre-specified significance and sensitivity analyses were inconsistent. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The pivotal study narratives align endpoint definitions, analysis populations, and outcome interpretation. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The trial reports connect efficacy conclusions to the prespecified analysis framework and observed safety findings. All referenced documents are available in the dossier annex and traceable by section identifier. Method validation records and quality controls were reviewed for internal consistency. The applicant confirms alignment with applicable technical guidance and regional submission standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Clinical study reports document protocol adherence, endpoint handling, and safety interpretation in the

[5] m5_bioequivalence - Biopharmaceutics and Bioequivalence Evidence
Labels: presence=present; length=length_ok; correctness=correct
Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of Cmax and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ($f2 > 50$). The bioanalytical methods used for plasma quantification were fully validated. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Statistical significance was set at an alpha level of 0.05, and all test